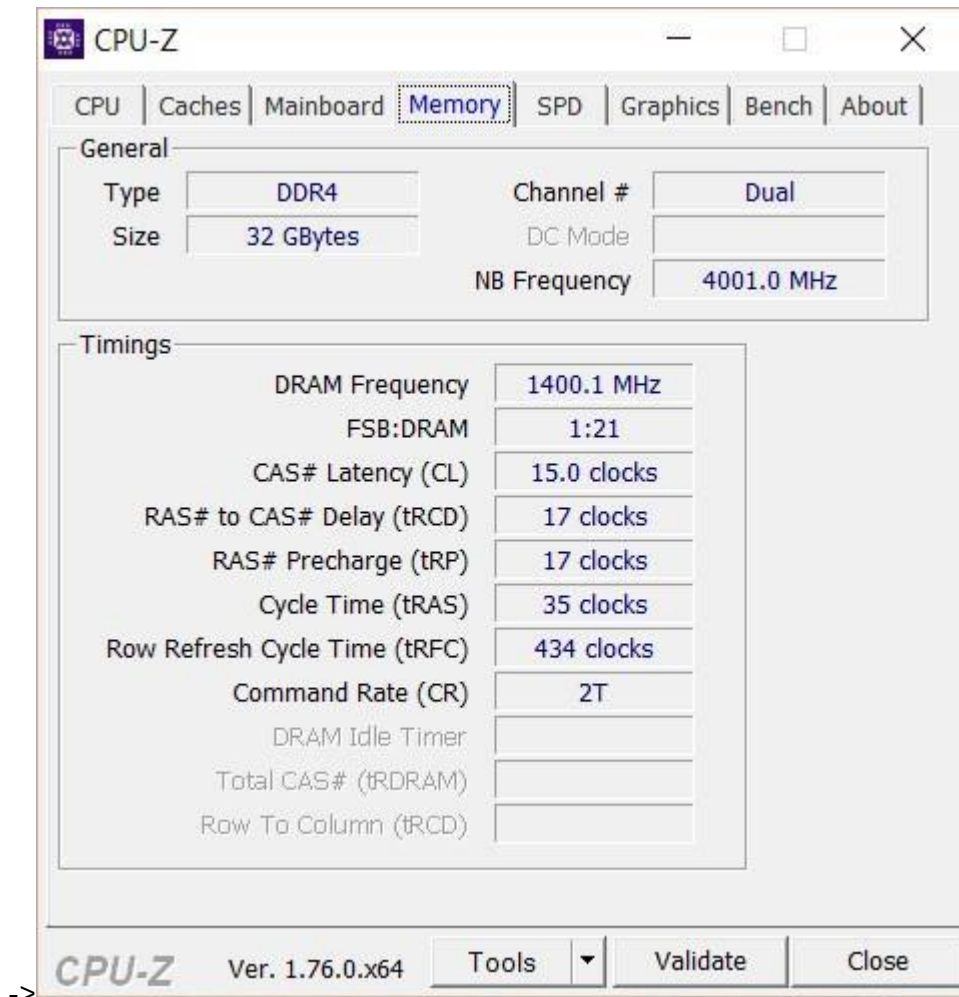


RAM AND STORAGE

1. Open your own laptop or desktop (or use a lab machine) and use CPU-Z to find out the exact RAM type (DDR3, DDR4, or DDR5), frequency (in MHz), and total installed memory. Take a screenshot of the Memory tab and note down these details



Open **CPU-Z** → **Memory** and record:

CPU-Z Detail	Example
Type	DDR4
DRAM Frequency	1600 MHz
Total Installed Memory	8 GB
Channel #	Dual

CPU-Z Detail	Example
Screenshot	Attach your CPU-Z Memory tab

Note: If CPU-Z shows **1600 MHz DRAM Frequency**, DDR memory effectively transfers data at about **3200 MT/s** (DDR = Double Data Rate). So don't confuse the CPU-Z frequency with the advertised DDR speed.

2. Physically examine a RAM module and a storage drive (HDD or SSD) available to you. Identify and write down the form factor (e.g., DIMM, SO-DIMM for RAM; 2.5-inch, M.2 for storage) and the connector type (e.g., SATA, M.2, PCIe) for each.

->RAM

Feature	What to identify
Form factor	DIMM — desktop / SO-DIMM — laptop
Connector	Memory module edge connector
Type	DDR3 / DDR4 / DDR5
Typical location	RAM slot on motherboard

Example:

Desktop DDR4 RAM → **DIMM** form factor.

Laptop DDR4 RAM → **SO-DIMM** form factor.

Storage

Storage Drive	Form Factor	Connector / Interface
2.5-inch SATA SSD	2.5-inch	SATA
2.5-inch HDD	2.5-inch	SATA
M.2 NVMe SSD	M.2	PCIe/NVMe

Example:

An M.2 NVMe SSD is a small circuit-board-style drive that plugs directly into an **M.2 slot** on the motherboard and commonly communicates using **PCIe/NVMe**.

3. Create a comparison table with at least 4 points contrasting HDD and SSD based on speed, durability, cost, and typical use cases. For each point, give a real-world example (e.g., 'SSD: Used in gaming laptops for fast load times').

->

Point	HDD	SSD
Speed	Slower because it uses spinning magnetic disks	Much faster because it uses flash memory
Durability	More vulnerable to physical shock because it contains moving parts	More resistant to shock because it has no moving mechanical parts
Cost	Usually cheaper per GB, especially for large capacities	Usually costs more per GB
Typical use	Large backups, archives, bulk storage	Operating systems, applications, gaming and laptops
Real-world example	A 4TB HDD used to store photos and video backups	An NVMe SSD used in a gaming PC for fast game loading

Simple comparison

HDD: Best when you need **large capacity at lower cost**.

SSD: Best when you need **speed, responsiveness and durability**.

4. Calculate the minimum storage capacity needed for each of these scenarios: (a) a YouTuber editing 4K videos weekly, (b) a programmer with lots of code projects and IDEs, (c) a gamer who wants to install 5 AAA games (assume each game is 100GB). Show your calculations.

->(a) **YouTuber editing 4K videos weekly**

4K video files can be very large. For a practical assignment calculation, assume:

- 4K raw/working footage per week = **200 GB**
- Keep 4 weeks of footage = 200×4
- Editing/project files and exports = **200 GB**
- Applications and operating system = **100 GB**

Calculation:

$$200 \times 4 = \mathbf{800\ GB}$$

$$800 + 200 + 100 = \mathbf{1,100\ GB}$$

Minimum recommended capacity: 1.1 TB

A practical choice would be a **2 TB NVMe SSD**, because it provides additional working space for future projects.

(b) Programmer with many projects and IDEs

Assume:

- Code projects = **100 GB**
- IDEs and development tools = **100 GB**
- SDKs/emulators/dependencies = **150 GB**
- Operating system and applications = **100 GB**
- Extra working space = **150 GB**

Calculation:

$$100 + 100 + 150 + 100 + 150 = \mathbf{600\ GB}$$

Minimum recommended capacity: 600 GB

A practical choice would be a **1 TB SSD**, preferably NVMe.

(c) Gamer installing 5 AAA games

Assumption given:

- Each game = **100 GB**
- Number of games = **5**

Calculation:

$$5 \times 100 \text{ GB} = \mathbf{500 \text{ GB}}$$

Minimum required capacity: 500 GB

However, the operating system, game launchers, updates and other files also require space.

Therefore, a **1 TB SSD** would be a more practical choice.

5. Based on your research and calculations, recommend the best storage solution (type and interface) for these three users: (1) a college student using only MS Office and web browsing, (2) a mobile app developer running Android Studio, (3) a competitive gamer who streams on Twitch. Justify your choice for each.

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User	Recommended Storage	Interface	Reason
College student — MS Office + web browsing	500 GB SSD	SATA or NVMe	Fast booting and application loading; 500 GB is sufficient for documents, Office and normal browsing
Mobile app developer — Android Studio	1 TB NVMe SSD	PCIe/NVMe	Android Studio, SDKs, emulators and project files can consume significant storage; NVMe provides fast application and file access
Competitive gamer + Twitch streaming	2 TB NVMe SSD	PCIe/NVMe	Provides space for large games and recordings while offering fast game loading and quick access to streaming files